Software Requirement Specification (SRS)
Physiological Digital Twin Berbasis CAPAR

Dokumen ini mendefinisikan kebutuhan perangkat lunak untuk membangun Physiological Digital Twin berbasis CAPAR. Sistem mengintegrasikan data wearable, konteks aktivitas, informasi klinis, model fisiologi, state estimation, dan explainable decision support.

1. Tujuan Sistem

Sistem harus mampu mengubah observasi fisiologis longitudinal menjadi estimasi resilience state, CAPAR event-state, trajectory recovery, phenotype regulasi, dan XAI.

2. Ruang Lingkup

Sistem mencakup akuisisi wearable (HR, RR/IBI, HRV, DFA, ACC), input konteks (aktivitas, tidur, stres, obat), input klinis (risk factor, examination, laboratory), CAPAR engine, resilience estimation, visualization, dan reporting.

3. Model Input Sistem

u(k): observable input berupa aktivitas, konteks, stres, tidur, dan obat.
d(k): disturbance berupa noise, unobserved physiological load, dan lingkungan.
c(k): clinical input berupa risk factor, examination, dan laboratory.
y(k): observation berupa HR, RR, HRV, DFA, dan sensor wearable.

4. Arsitektur Sistem

```
Clinical Input c(k)
        |
Wearable + Context u(k)
        |
Disturbance d(k)
        |
        v
+----------------+
| CAPAR ENGINE   |
+----------------+
        |
        v
Resilience State X(k)
        |
        v
CAPAR Event State s(k)
        |
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```

```
 |       |         |       |
Risk  Recovery  Phenotype  XAI
```

5. Functional Requirement

FR-01 Data acquisition:
Sistem menerima data wearable dengan timestamp.

FR-02 Context engine:
Sistem mengelola aktivitas, tidur, stres, obat.

FR-03 Clinical engine:
Sistem menyimpan clinical prior.

FR-04 Quality gate:
Sistem mengevaluasi kualitas sinyal.

FR-05 Baseline engine:
Sistem membuat personal-contextual baseline.

FR-06 Feature engine:
Menghitung HR, HRV, DFA, persistence, recovery, relapse.

FR-07 CAPAR FSM:
Menghasilkan state transition.

FR-08 Resilience engine:
Menghitung Vulnerability, Cardiac Reserve, Autonomic Reserve, Recovery Capacity, Regulation Stability.

FR-09 XAI:
Menampilkan evidence pendukung, kontra evidence, dan uncertainty.

6. CAPAR State Requirement

State yang harus didukung:

1. Baseline Mature
2. No Deviation
3. Deviation
4. Persistent Deviation
5. Recovery Start
6. Full Recovery
7. Relapse

8. Baseline Update

Setiap state harus memiliki:
- definisi fisiologis,
- kondisi masuk,
- kondisi keluar,
- parameter monitoring.

7. Mathematical Requirement

State space model:

$$z(k+1)=f(z(k),u(k),d(k))+w(k)$$

$$y(k)=h(z(k),u(k))+v(k)$$

$$X(k)=g(z(k),c(k),Baseline,e(k))$$

$$s(k+1)=T(s(k),D(k),P(k),R(k))$$

$z(k)$ = latent physiological state
$X(k)$ = resilience state
$s(k)$ = CAPAR event state

8. Resilience Output

Output utama:

1. Vulnerability/Risk
2. Recovery Trajectory
3. Regulation Phenotype
4. Decision Support
5. Explainable AI

Setiap output harus memiliki confidence dan data lineage.

9. Software Module Requirement

Modul perangkat lunak:

/data
- wearable ingestion

- clinical database
- context database

/preprocessing
- filtering
- artifact removal
- quality scoring

/features
- HR features
- HRV features
- DFA features
- dynamic features

/engine
- baseline engine
- deviation detector
- recovery engine
- relapse detector
- FSM controller

/xai
- explanation generator

/ui
- dashboard
- trajectory viewer
- report generator

10. Non Functional Requirement

Reliability:
Sistem harus mampu monitoring longitudinal.

Security:
Data kesehatan harus terlindungi.

Traceability:
Output harus dapat ditelusuri dari sumber data.

Explainability:
Setiap keputusan harus memiliki alasan.

Maintainability:
Rule, model, dan threshold dapat diperbarui.

11. Validation Requirement

Validasi dilakukan pada:

1. Signal level:
ECG/reference device.

2. Feature level:
Agreement analysis.

3. Episode level:
EMA/event annotation, F1 score.

4. Trajectory level:
TTR error dan recovery similarity.

5. Phenotype level:
Repeatability dan stability.

12. Acceptance Criteria

Sistem diterima apabila:

- mampu menghasilkan CAPAR event-state;
- mampu menghitung resilience domain;
- mampu menampilkan recovery trajectory;
- mampu memberikan XAI;
- memiliki provenance dari input sampai output.

13. Peran Konseptor dan Programmer

Konseptor bertanggung jawab terhadap semantic model, definisi fisiologi, hubungan feature-latent variable, validasi, dan governance. Programmer bertanggung jawab terhadap implementasi pipeline data, algoritma, database, API, UI, dan deployment.